Meeting 5C — Input Layer and Feature Layer Software Architecture

Digital Harmonium Workstation

Final Design Specification

Purpose of these Layers

These two layers form the bridge between the musician and the internal audio engine.

The **Input Layer** is responsible for observing physical hardware and converting user interactions into clean software events.

The **Feature Layer** receives those events and decides what the workstation should do in response.

Neither layer generates audio.

Neither layer performs DSP.

They simply coordinate the entire workstation.

Layer 5 — Input Layer

Philosophy

The Input Layer converts electrical signals into meaningful software events.

Instead of allowing every subsystem to directly read hardware, all physical devices are managed through dedicated modules.

This creates a clean separation between hardware and software.

Module 1 — Keyboard Scanner

Purpose

Continuously scans the repaired keyboard matrix and converts switch closures into note events.

Responsibilities

- Scan keyboard matrix continuously
- Detect key press
- Detect key release
- Debounce keys
- Detect multiple simultaneous keys
- Generate note events
- Report key velocity timing (future expansion)
- Forward clean events to Runtime Layer

Inputs

- Keyboard matrix

Outputs

- KeyPressed
- KeyReleased

Does NOT

- Play audio
- Allocate voices
- Generate MIDI
- Manage instruments

Module 2 — Button Manager

Purpose

Manages all dedicated control buttons used throughout the workstation.

Responsibilities

- Scan buttons
- Debounce switches
- Detect short press
- Support future long press functionality
- Generate button events

Controlled Buttons

- Voice
- Tabla
- Tanpura
- Metronome
- Record
- Playback
- Menu
- Back
- Start
- Stop
- Future buttons

Outputs

- ButtonPressed
- ButtonReleased

Does NOT

- Change menus
- Start recording
- Change instruments

It only reports button activity.

Module 3 — Slider Manager

Purpose

Handles analog sliders available on the keyboard.

Responsibilities

- Read ADC values
- Smooth noisy signals
- Scale values
- Detect movement
- Generate slider events

Planned Sliders

Slider 1

- Master Volume

Slider 2

- Assignable
- LCD Contrast
- Reverb
- Expression
- Future parameters

Outputs

- SliderChanged

Does NOT

Directly modify audio.

Module 4 — LED Manager

Purpose

Controls every LED available on the keyboard.

This transforms the repaired keyboard into an educational workstation.

Responsibilities

- Boot LED test
- Playback guidance
- Recording guidance
- Status indication
- Animation effects
- Error indication
- Future learning modes

LED Uses

- Recording indicator
- Playback indicator
- Learning mode
- Startup animation
- Error display
- Future note guidance

Outputs

LED control only.

Does NOT

Interpret music.

Module 5 — Display Manager

Purpose

Owns the complete 16×2 LCD display.

No other module writes directly to the display.

Responsibilities

- Draw screens
- Refresh display
- Manage menus

- Update status indicators
- Display current parameters
- Draw boot screen
- Draw error messages

Displays

- Home screen
- Instrument screens
- Tabla screen
- Tanpura screen
- Recording screen
- Playback screen
- Settings screen

Does NOT

Control features directly.

Input Layer Summary

The Input Layer never makes decisions.

It simply reports:

- Keys
- Buttons
- Sliders
- LEDs
- Display

to higher software layers.

Layer 6 — Feature Layer

Philosophy

This layer answers one question:

> "Given what the musician just did, what should the workstation do?"

It coordinates features without directly touching hardware or generating audio.

Module 6 — Voice Selection Manager

Purpose

Controls active instrument selection.

Responsibilities

- Cycle instruments
- Load requested instrument
- Remember current selection
- Notify Runtime Layer
- Trigger VHL loading when required

Controls

- Harmonium
- Organ
- Future instruments

Does NOT

Generate sound.

Module 7 — Accompaniment Manager

Purpose

Coordinates all accompaniment engines.

Responsibilities

- Enable accompaniment
- Disable accompaniment
- Prevent conflicts
- Coordinate multiple accompaniment sources

- Synchronize timing

Managed Systems

- Tabla
- Tanpura
- Metronome

Module 8 — Tabla Feature

Purpose

Provides complete rhythm accompaniment.

Responsibilities

- Select taal
- Start playback
- Stop playback
- Adjust tempo
- Adjust transpose
- Synchronize looping
- Display current rhythm

Supported Taals

- Tritaal
- Ektaal
- Jhaptaal
- Chautaal
- Rupak
- Dadra
- Future taals

Does NOT

Mix audio.

Module 9 — Tanpura Feature

Purpose

Provides continuous drone accompaniment.

Responsibilities

- Start drone
- Stop drone
- Change tonic
- Adjust tempo (if applicable)
- Adjust volume
- Display drone status

Does NOT

Generate notes manually.

Module 10 — Metronome Feature

Purpose

Provides timing assistance.

Responsibilities

- Start metronome
- Stop metronome
- Change BPM
- Accent beats
- Select time signature

Future

Support multiple beat patterns.

Module 11 — Recording Manager

Purpose

Records musical performance.

Responsibilities

- Start recording
- Stop recording
- Store note events
- Store timestamps
- Store velocity
- Save performance

Recording Type

Performance recording

NOT audio recording.

Future

Save recordings to SD Card.

Module 12 — Playback Manager

Purpose

Replays recorded performances.

Responsibilities

- Load recording
- Replay notes
- Replay timing
- Replay velocities
- Synchronize LEDs
- Synchronize accompaniment

Does NOT

Modify recordings.

Module 13 — Transpose Manager

Purpose

Implements global instrument transposition.

Responsibilities

- Shift tonic
- Shift note mapping
- Apply transpose to active instrument
- Coordinate accompaniment transpose

Importance

Critical for Indian Classical Music where Sa changes frequently.

Module 14 — Settings Manager

Purpose

Stores workstation configuration.

Responsibilities

- Speaker volume
- LCD contrast
- Startup instrument
- Default transpose
- User preferences
- Future configuration

Future

Persistent settings stored on SD Card.

Module 15 — Storage Manager

Purpose

Handles long-term storage.

Responsibilities

- Load VHL files
- Save recordings
- Read recordings
- Manage configuration files
- Organize SD Card data

Current Philosophy

VHL instruments remain on the SD card.

When the user selects an instrument:

1. Read VHL
2. Load into RAM
3. Use until another instrument is selected
4. Replace previous instrument

This allows virtually unlimited instrument expansion without exhausting ESP32 RAM.

Feature Layer Summary

The Feature Layer never scans hardware.

The Feature Layer never produces sound.

It simply coordinates workstation behaviour by deciding **what** should happen after each user action.

Overall Workflow

```
Hardware

↓

Input Layer

↓

Runtime Layer

↓

Feature Layer

↓

Instrument Layer

↓

Audio Layer

↓

Speaker
```

Each layer has a single, well-defined responsibility, making the software modular, maintainable, and easy to extend in future revisions.

End of Meeting 5C — Input Layer and Feature Layer Architecture